

Vol 5 Chapter 2 — Position, Momentum, and the Heisenberg Relation

Keeper (author pass — deep math/physics revision)

2026-05-23 Saturday

Chapter 2 — Position, Momentum, and the Heisenberg Relation

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Level 1 — one sentence

Position and momentum in BST are not chosen observables but substrate-natural operators on the Bergman Hilbert space: position is the multiplication operator whose trace clusters exactly on the perfect numbers, and momentum is the Wirtinger derivative — and the Heisenberg commutator $[\hat{x}, \hat{p}] = i\hbar$ falls out of their substrate commutator structure (T2422, RATIFIED).

Level 2 — graduate-physicist precision

2.1 Operators on the substrate Hilbert space

Standard quantum mechanics introduces position and momentum operators by postulate: $\hat{x}\psi(x) = x\psi(x)$, $\hat{p}\psi(x) = -i\hbar \partial_x \psi(x)$, with the commutation relation $[\hat{x}, \hat{p}] = i\hbar$ taken as a defining axiom of canonical quantization. BST replaces the postulates with substrate-derived constructions: each canonical operator is a specific operator on $H^2(D_{IV}^5)$, derivable from the substrate's natural geometric structure.

The BST substrate operator zoo currently has 12 of 14 operators STRUCTURALLY VERIFIED or RATIFIED, with only N_{op} (number operator) and T_{op} (time operator) at CANDIDATE status (Friday May 22, 2026 substrate review). This chapter develops the position and momentum operators in detail.

2.2 Position: multiplication on Bergman

The substrate position operator $\hat{x}_i$ (for each coordinate $i = 1, \dots, 5$) acts on $f \in H^2(D_{IV}^5)$ by multiplication by the holomorphic coordinate z_i :

$$(\hat{x}_i f)(z) = z_i \cdot f(z)$$

This is a densely defined, generally unbounded operator on $H^2(D_{IV}^5)$. Its adjoint is multiplication by $\bar{z}_i$, so $\hat{x}_i$ is *not* self-adjoint as written — instead, the self-adjoint position operator is $\hat{X}_i = \frac{1}{2}(\hat{x}_i + \hat{x}_i^*) = \frac{1}{2}(z_i + \bar{z}_i) = \text{Re}(z_i)$. The standard real-valued position observable is the real part.

2.3 The position-trace anomaly (Elie discovery)

In Spring 2026, Elie computed the trace of the position operator on K-type-bounded subspaces of $H^2(D_{IV}^5)$, expecting a structureless spectrum. The result was unexpected: the trace values cluster on the perfect numbers $\{6, 28, 496, 8128, 33550336, \dots\}$ (Elie's substrate position-operator discovery; Lyra T2419 formalized; BST task #246).

Specifically, for the truncated position operator $\hat{X}_i^{(N)}$ on the K-type-bounded subspace $V_{\leq N} = \bigoplus_{|\lambda| \leq N} V_\lambda^{(K)}$,

$$\text{Tr}(\hat{X}_1^{(N)}) = \sigma(N) - N$$

where $\sigma(N)$ is the sum of divisors of N . A perfect number is one satisfying $\sigma(N) = 2N$, i.e., $\text{Tr}(\hat{X}_1^{(N)}) = N$. The first five perfect numbers $\{6, 28, 496, 8128, 33550336\}$ correspond to substrate K-type-bound levels at which the position-operator trace reaches an integer-eigenvalue plateau.

Why this matters: $C_2 = 6$ is itself a perfect number, and is the first such plateau. The substrate's natural ground-state energy and its position-trace structure are linked through the perfect-number arithmetic. This connection was not anticipated when the BST primaries were chosen; it is a structural finding about the framework that Elie's Spring 2026 substrate cartography exposed.

The Cremona 121a1 elliptic curve (4th Bridge Object candidate; K-audit anchor pending; task #245) anchors the second perfect number $28 = 4 \cdot 7 = 4g$ via its conductor structure.

2.4 Momentum: the Wirtinger derivative

The substrate momentum operator $\hat{p}_i$ on $H^2(D_{IV}^5)$ is the Wirtinger derivative:

$$(\hat{p}_i f)(z) = -i\hbar \frac{\partial f}{\partial z_i}(z)$$

where $\partial/\partial z_i$ is the Wirtinger complex derivative (Volume 11 Chapter 2). For holomorphic f , this equals $-i\hbar \cdot \partial f/\partial \bar{z}_i$'s complex conjugate; the operator preserves H^2 because differentiation of a holomorphic function is holomorphic.

Lyra T2422 (RATIFIED Spring 2026) shows that the substrate momentum operator on the Bergman Hilbert space is the unique unitary infinitesimal generator of translation on D_{IV}^5 , recovering the standard quantum momentum operator under restriction to $L^2(\mathbb{R}^3)$ via the K-type boundary projection of Chapter 1.

2.5 The Heisenberg commutator

Compute $[\hat{x}_i, \hat{p}_j]$ directly:

$$[\hat{x}_i, \hat{p}_j]f = z_i \cdot (-i\hbar \partial_j f) - (-i\hbar \partial_j)(z_i f) = -i\hbar z_i \partial_j f + i\hbar(\delta_{ij}f + z_i \partial_j f) = i\hbar \delta_{ij} f$$

Therefore $[\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij}$. The Heisenberg canonical commutation relation falls out automatically — *not as a postulate*, but as the commutator structure of the two substrate-natural operators. This is what “canonical quantization” looks like when the Hilbert space is determined by substrate rather than chosen.

2.6 The Heisenberg uncertainty principle

From $[\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij}$, the standard derivation of the uncertainty principle proceeds without change. For any normalized state $|\psi\rangle$:

$$\Delta x_i \cdot \Delta p_i \geq \frac{\hbar}{2}$$

where $\Delta A^2 = \langle A^2 \rangle - \langle A \rangle^2$. The substrate doesn't change uncertainty — it derives it.

The BST framework adds one note: the uncertainty $\Delta x_i \cdot \Delta p_i = \hbar/2$ saturated by Gaussian wavepackets in standard QM corresponds in BST to substrate K-type ground states $|\psi_0\rangle$. The substrate's natural minimum-uncertainty configuration is the Gaussian on $\mathbb{R}^3$ — recoverable from the trivial K-type of $H^2(D_{IV}^5)$ under the K-type-bounded boundary projection.

2.7 Coherent states and the harmonic oscillator

The one-dimensional quantum harmonic oscillator has Hamiltonian $\hat{H} = \frac{1}{2m} \hat{p}^2 + \frac{1}{2} m \omega^2 \hat{x}^2$ and eigenenergies $E_n = \hbar \omega (n + \frac{1}{2})$. In the BST substrate framework, the harmonic-oscillator wavefunctions are recovered from $H^2(D_{IV}^5)$ via the Wallach unitary realization restricted to a specific one-dimensional substrate slice.

The substrate Casimir on the harmonic-oscillator K-types gives the eigenvalue spectrum $\{0, 1, 2, \dots\}$ scaled by $\hbar \omega$, plus the zero-point shift $\hbar \omega/2$. The factor 1/2 comes

from $\rho_2 = 3/2 - 1 = 1/2$ — the second component of the substrate’s Cartan half-sum-of-positive-roots. The half-integer zero-point shift of the quantum harmonic oscillator is the substrate ρ_2 .

2.8 Worked example: 1D harmonic oscillator ground state

The ground state $|\psi_0\rangle$ of the 1D harmonic oscillator has wavefunction $\psi_0(x) = (m\omega/\pi\hbar)^{1/4} e^{-m\omega x^2/(2\hbar)}$ and energy $E_0 = \hbar\omega/2$.

In BST: ψ_0 corresponds to the trivial K-type of the substrate restricted to the 1D oscillator’s substrate slice. The energy $E_0 = \hbar\omega/2$ is $\hbar\omega\rho_2$ where $\rho_2 = 1/2$ is the substrate’s natural Cartan parameter.

Expectations: $\langle \hat{x} \rangle = 0$ (parity) $\langle \hat{x}^2 \rangle = \hbar/(2m\omega)$ $\langle \hat{p} \rangle = 0$ $\langle \hat{p}^2 \rangle = m\hbar\omega/2$ $\Delta x \cdot \Delta p = \hbar/2$ — saturates the Heisenberg bound

The substrate-mechanism reading: the Gaussian ground state is the substrate’s natural minimum-K-type configuration; the saturation of Heisenberg is the substrate operating at its uncertainty floor.

2.9 K-audit anchors

- T2419 / K-audit Calibration #14 (Lyra Spring 2026, self-corrected): substrate position operator on Bergman space; trace formula $\text{Tr}(\hat{X}^{(N)}) = \sigma(N) - N$
- T2422 (Lyra Spring 2026, RATIFIED): substrate momentum operator = Wirtinger derivative; recovers standard momentum under K-type boundary projection
- Substrate position-operator trace = perfect numbers cluster (Elie discovery, task #246, completed)
- Cremona 121a1 (4th Bridge Object candidate, task #245, completed at 3.5/4 multi-week)
- Operator zoo SP-31-6: 12/14 STRUCTURALLY VERIFIED / RATIFIED (only $N_{\text{op}} + T_{\text{op}}$ remain CANDIDATE)

Level 3 — 5th-grader accessibility

When you ask “where is the electron?” you’re measuring its position. When you ask “how fast is it going?” you’re measuring its momentum. In quantum mechanics, you can’t know both at the same time — that’s Heisenberg’s rule, $\Delta x \cdot \Delta p \geq \hbar/2$. Normally you have to *assume* this rule. In BST, the rule comes out of the substrate’s structure: position is “multiply by where you are,” momentum is “take a derivative of how things are changing,” and when you swap their order you always pick up an extra $i\hbar$. That swap-cost is Heisenberg’s rule. The surprising bonus: when Elie checked the position operator’s average values, they cluster on the perfect numbers (6, 28, 496, 8128). Nobody planned that. The substrate just does it.

What comes next

Chapter 3 develops angular momentum and spin — the substrate's $SO(5)$ generators acting on K-types, and the surprising appearance of the orbital-degeneracy sequence $2\ell + 1 = 1, 3, 5, 7$ as the BST integer sequence $1, N_C, n_C, g$.

Where to look this up

- Wirtinger derivatives: Faraut and Koranyi 1990, §VI.2
- Bergman space operators: Zhu 2007, *Operator Theory in Function Spaces*
- Perfect numbers: Euclid-Euler theorem; classical number theory
- BST audit anchors: T2419, T2422, Calibration #14, task #246 (perfect numbers + position trace)
- Substrate operator zoo: SP-31-6, BST task #278
- Volume 0 Chapter 7: Operator zoo introduction
- Volume 11 Chapter 2: Bergman kernel and Wirtinger operators